DES423 · PREDICTIVE MODELING PITCH

SessionSignal

Can browsing behavior tell us which shopping sessions will end in a purchase?

UCI Online Shoppers Purchasing Intention · 12,330 sessions · live demonstration · 7-member team

PROBLEM IMPORTANCE

Lost sessions are a large, persistent cost

$6.42T

global retail ecommerce sales in 2025 — 20.5% of all retail

70.22%

average documented cart abandonment, 50 studies 2006–2025

84.5%

of sessions in our dataset ended without a purchase

Abandonment has sat near 70% for two decades, so this is a **structural gap, not a temporary dip**. At a $6.42 trillion market scale, even small conversion gains are commercially significant.

No store can hand-review every lost session. **Analyst attention is the scarce resource** — and that is what we target.

Sources: eMarketer 2025 forecast; Baymard Institute (unweighted average of 50 studies, range 55%–84%); UCI dataset 468. Cart abandonment and session non-purchase are related but different measures.

PROBLEM STATEMENT

Which lost sessions deserve attention first?

User: an ecommerce conversion analyst.

Situation: thousands of sessions end without a purchase; only a handful can ever be investigated.

Question: which ones should the team examine first?

Objective: rank completed sessions by purchase probability, using browsing behavior alone.

How we measure success

Concentrate real buyers into the top-scored slice. Our primary metric is **average precision**, because only 15.5% of sessions convert — plain accuracy would reward always predicting “no purchase”.

Explicitly out of scope

Why a shopper left · guaranteed revenue recovery · real-time intervention mid-session.

DATASET AND TARGET

One row = one shopping session

12,330
historical sessions

17
input features in the original data

15.5%
ended in a purchase

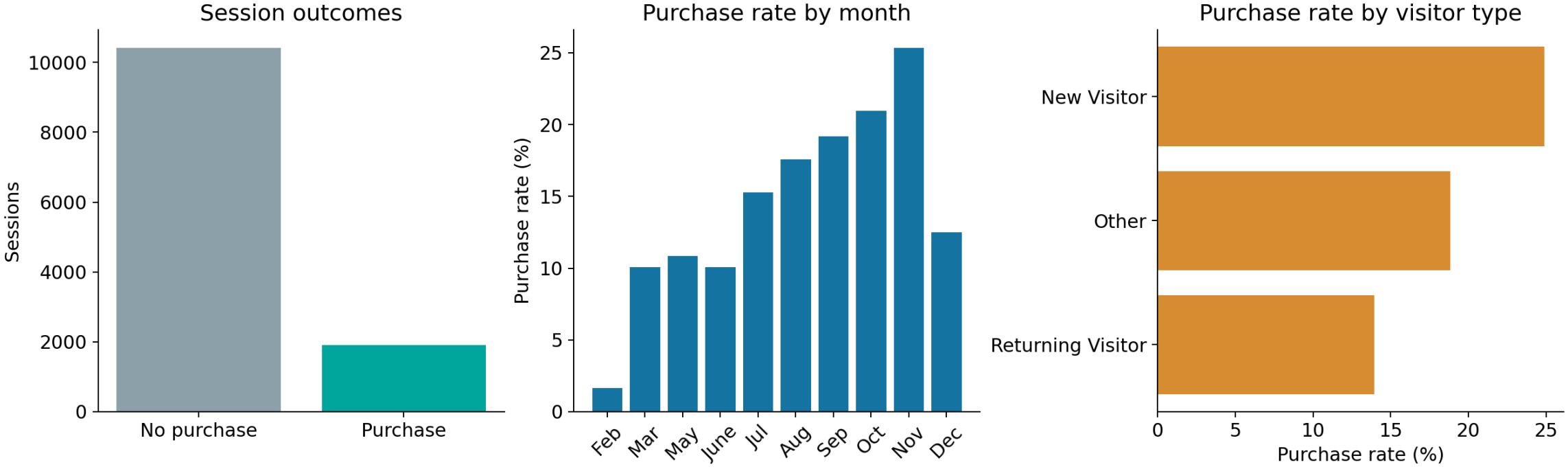
Predict: Revenue = purchase or no purchase.

Inputs: page counts and time, analytics rates, visitor type, month, device and traffic categories.

Source: UCI Machine Learning Repository, dataset 468

EXPLORATORY ANALYSIS

Purchases vary by month and visitor type



New visitors purchased in 24.9% of sessions versus 13.9% for returning visitors. November reached 25.4%; February was 1.6%.

Calculated from the supplied UCI CSV. These are associations, not causal effects. Month groups have different sample sizes.

EVALUATION DESIGN

Keep the test set unseen until the end

01 Remove 125 exact duplicate rows

02 Exclude `PageValues`, a revenue-derived analytics feature

03 Stratified 80/20 split: 9,764 train and 2,441 test sessions

04 Four-fold cross-validation and tuning on training data only

Primary measure: average precision (AP), because purchases are uncommon.

MODELS

Models for purchase classification

Logistic regression

Combines feature values into a linear score. Class weighting gives rare purchases more influence during training.

Random forest

Averages predictions from many decision trees to capture nonlinear patterns and interactions.

Gradient boosting

Adds trees sequentially to improve earlier predictions. The main pipeline uses histogram gradient boosting.

No-purchase baseline

Predicts no purchase for every session. Its high accuracy shows why purchase recall also matters.

PYCARET CLASSIFICATION

A low-code model comparison

Companion workflow

Same cleaned CSV and target, with `PageValues` excluded.

1. `setup`
   80/20 split and four folds
2. `compare_models`
   Compare three models by AUC
3. `predict_model`
   Predict on PyCaret's holdout

Saved four-fold CV AUC

Model	AUC
Gradient boosting	0.7703
Random forest	0.7596
Logistic regression	0.7098

PyCaret uses its own split and model settings. Read these CV scores separately from the main test results.

The app uses the tuned scikit-learn forest. PyCaret demonstrates the tutorial workflow.

Saved run: PyCaret leaderboard. Code: companion script. Reference: PyCaret tutorials.

HELD-OUT RESULTS

The tuned forest leads on average precision



Selection used training-fold AP. Test ROC AUC: **0.787**.

Tuning searched depths {8, 16, None} and leaf sizes {2, 5}. Best: depth None, leaf size 5. CV AP rose from **0.386 to 0.391**.

TRADEOFFS

Ranking is useful; a 0.5 cutoff misses buyers

Model	Precision	Recall	F1
Logistic regression	27.9%	78.3%	0.411
Random forest	39.9%	34.6%	0.370
Gradient boosting	45.8%	7.1%	0.122
Forest, tuned	**37.1%**	**46.3%**	**0.412**

The top 10% of forest scores contain **43.7%** purchasers, about **2.79×** the overall test rate.

All values use the same untouched 2,441-session test set; threshold metrics use 0.5.

PRODUCT CONCEPT

Choosing the next conversion experiment

Proposed analyst workflow

1. Rank completed sessions and review high scores with no purchase.
2. Inspect shared patterns and available journey or support logs.
3. Choose one hypothesis for a controlled experiment.

Why ranking pays off

2.79×

more buyers per session reviewed. The top-scored 10% is **43.7%** buyers against a **15.6%** base rate.

The same analyst hour surfaces nearly three times as many relevant sessions as reviewing at random.

Journey logs would add evidence. The score alone does not explain why a shopper left.

Current demo: score individual sessions. Batch ranking is a proposed extension. An A/B test would measure any sales impact.

LIVE DEMONSTRATION

Try a held-out session

```
Example 27 → score 0.68 · actually bought  ✓ caught a buyer
Example  9 → score 0.01 · did not buy  ✓ correctly ignored
Example  8 → score 0.64 · did not buy  ← investigate this one
```

All three are held-out sessions the model never trained on. Example 8 is the point of the product: high intent, no purchase — exactly the session an analyst should open first.

NEXT STEPS

Make the model more trustworthy

- Collect timestamped partial sessions for genuine live prediction.
- Validate on a later time period and another store.
- Check calibration, subgroup performance, and false positives.
- Choose an action threshold based on cost and capacity.
- Run an A/B test before claiming sales impact.

TEAM AND CONTRIBUTIONS

Who did what

Student ID	Name	Main contribution
6622782298	Sirawich Narakulsookpipat	Problem research and framing
6622780342	Chaitawat Boonkitticharoen	Dataset audit and target definition
6622782215	Bhuchi Kumprakob	Exploratory data analysis
6622780276	Nonthaphat Chuathong	Evaluation design and model selection
6622770657	Kamin Sirisuwapong	PyCaret comparison and hyperparameter tuning
6622772455	Poonnawit Wongprateepwilai	Results analysis and business solution
6622790267	Choltawat Jullakasevi	Streamlit app, demonstration and limitations

Every member speaks to their own section in this video.

TAKEAWAY

We can rank purchase intent. We still need to test the intervention.

The tuned forest improves AP from 0.156 to 0.374 on held-out sessions.

Sirawich · Chaitawat · Bhuchi · Nonthaphat · Kamin · Poonnawit · Cholwat